

$$\begin{array}{ccccc}
R^2G(M, E) & \xrightarrow{\mu_{G(M, E)}^R} & & & RG(M, E) \\
R\chi_E^M \downarrow & & & & \downarrow \chi_E^M \\
RG(M, TE) & \xrightarrow{\chi_{TE}^M} & G(M, T^2E) & \xrightarrow{G(M, \mu_E^T)} & G(M, TE)
\end{array}$$